James Ratan Dukkipati

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SUMMARY

Computer Science graduate student at Binghamton University (SUNY) specializing in Artificial Intelligence, with formal coursework in Computer Vision (Grade: A) and hands-on experience building and deploying AI models using **PyTorch** and **Python**. Trained deep learning pipelines covering dataset construction, model evaluation, and inference optimization. Experienced with **transformer-based** models (**SBERT**), **OpenCV**, image processing, and **FAISS**-based similarity search. Additional background in a government defense environment with hands-on systems and network operations. Available for Summer 2026; applying for the AI Vision Engineer internship.

EDUCATION

Binghamton University, State University of New York **Aug. 2024 – Dec. 2026**
M.S. Computer Science, Artificial Intelligence Track | Thomas J. Watson College | GPA: 3.7/4.0
Relevant Coursework: Introduction to Computer Vision (A), Machine Learning, Advanced Algorithms

MVSR Engineering College (Osmania University) **Aug. 2020 – May 2024**
Bachelor of Engineering in Computer Science | GPA: 8.2/10.0

TECHNICAL SKILLS

Computer Vision & Deep Learning: PyTorch, OpenCV, CNNs, Image Processing, Object Detection, SBERT
ML & Data: scikit-learn, FAISS, pgvector, TF-IDF, Hugging Face, Pandas, NumPy, Dataset Construction
AI & LLMs: LangChain, OpenAI API (GPT-4, Ada), RAG, Pinecone, Prompt Engineering
Languages: Python, Bash, TypeScript, JavaScript, SQL, Java
DevOps & Systems: Docker, Git, Linux, FastAPI, PostgreSQL, AWS, Vercel

PROFESSIONAL EXPERIENCE

Cybersecurity Intern **Jun. 2023 – Aug. 2023**
Defence Research and Development Organisation (DRDO) | Wazuh, SIEM, Linux, Python *Hyderabad, India*

- Embedded in a classified **government defense** environment supporting critical security infrastructure; deployed and maintained **Wazuh XDR/SIEM** across **50 endpoints** using **Linux** and **Python** automation to process **10,000+ events daily**.
- Implemented **CVE/CVSS** vulnerability detection and compliance monitoring across **3 frameworks** (**HIPAA, PCI DSS, GDPR**); mapped threat behaviors to **MITRE ATT&CK** for structured audit-ready reporting.
- Integrated **VirusTotal and YARA**-based detection into the automated pipeline; triaged **15+ security alerts** and authored runbooks reducing **MTTD** by **40%**.
- Collaborated with network engineers across **8 switches** to trace traffic anomalies and diagnose misconfigured **VLANs**; delivered remediation findings under strict operational timelines.

PROJECT EXPERIENCE

Game Recommendation System | *Python, PyTorch, SBERT, FAISS, pgvector, FastAPI, Docker* **Jan. 2024 – May 2024**

- Trained a deep learning pipeline in **Python** using **SBERT** (transformer-based embeddings via **PyTorch**) to encode semantic features across **50,000+** records; combined with **TF-IDF** and **ALS** collaborative filtering to outperform single-method baselines by **18%** on **NDCG@10**.
- Built and cleaned the full **training dataset** from raw interaction data in **Python**; benchmarked **3 model variants** using **NDCG** and **precision@k** to validate the optimal approach across all evaluation cuts.
- Integrated model outputs into a **FastAPI** serving layer with **FAISS** and **pgvector** for sub-**100ms** inference; containerized the full stack with **Docker Compose** for consistent test and production environments.

Agentic Security Log Analyzer | *Python, LangChain, GPT-4, FastAPI, pgvector, Docker* **Jan. 2025 – Present**

- Built an **AI pipeline** in **Python** using **LangChain** and **GPT-4** to autonomously classify raw inputs across **6 categories**; applied iterative **prompt engineering** to enforce structured **JSON** output and achieve reliable zero-touch escalation of high-severity findings.
- Designed **REST API** endpoints with **FastAPI** and **PostgreSQL** with **pgvector**; persisted **1536-dim** embeddings per record enabling sub-second semantic similarity search, reducing manual analysis time by **65%**.

DocConvo | *TypeScript, Next.js, LangChain, OpenAI (Ada/GPT-3.5), Pinecone, Stripe, Vercel* **Sep. 2024 – Dec. 2024**

- Built a production **RAG** platform integrating **LangChain** document chunking, **OpenAI Ada** embeddings, **Pinecone** vector retrieval, and **GPT-3.5 Turbo** generation; delivered sub-**3s** responses on files up to **100 pages** with real-time **WebSocket** streaming.
- Shipped the full stack as a **7-module monorepo** with authentication, **Stripe** billing, and file handling; deployed live on **Vercel** with full end-to-end functionality and active users.